INCH-POUND
MS27642G
20January 1998
SUPERSEDING
MS27642F
21 June 1995

MILITARY SPECIFICATION SHEET

BEARING, BALL, AIRFRAME, EXTRA LIGHT DUTY

This Specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-B-7949.

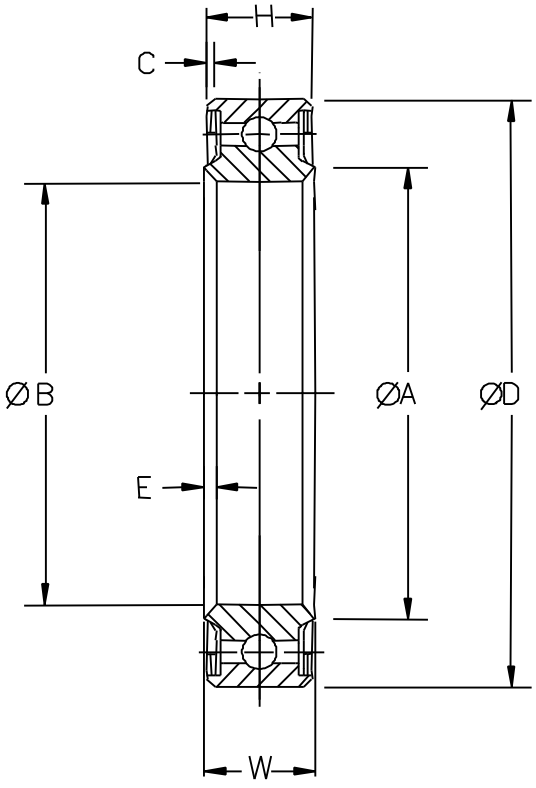

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TABLE I. BEARING DIMENSIONS AND MAXIMUM TORQUE

MS DASH NO.	∅ B 3/ BORE	BORE TOLERANCE WITHOUT SUFFIX S	BORE TOLERANCE WITH SUFFIX S	∅ D 2/ 3/ OUTSIDE DIAMETER +.0000 -.0010	W 2/ WIDTH INNER RING +.000 -.005	H WIDTH OUTER RING +.000 -.005	∅ A SHOULDER DIAMETER INNER RING (APPROX)	E AND C 4/ 5/ INNER & OUTER RING CORNER CHAMFER x 45° +.015 -.000	RADIAL INTERNAL CLEARANCE WITHOUT SUFFIX S	RADIAL INTERNAL CLEARANCE WITH SUFFIX S	WEIGHT POUNDS (APPROX)	MAXIMUM STARITNG TORQUE (INCH-OZ)
-16	1.0000	+.0000 -.0005		1.7500			1.141				.14	5
-21	1.3130			2.0625			1.454		.0003 TO .0010		.16	7
-23	1.4380			2.1875	.437	.375	1.575	.024			.17	8
-25	1.5630			2.3125			1.693				.19	9
-29	1.8130			2.5625			1.931				.21	12
-33	2.0630			2.8125			2.231				.23	14
-37	2.3130			3.0625			2.468				.26	17
-47	2.9380		+.0010 -.0005	3.8750			3.093			.0001 TO .0005	.49	24
-49	3.0630			4.0000			3.222				.53	25
-52	3.2500			4.1875			3.479				.55	26
-56	3.5000	+.0000 -.0005		4.4375	.531	.469	3.775				.58	28
-60	3.7500			4.6875			4.014				.61	28
-64	4.0000			4.9375			4.253	.039	.0003 TO .0015		.64	1/
-68	4.2500			5.3125			4.517				.73	
-72	4.5000			5.5625			4.774				.76	
-76	4.7500			5.8125			5.046				1.00	
-80	5.0000			6.0625	.593	.531	5.246				1.04	
-84	5.2500			6.3125			5.506				1.09	
-88	5.5000			6.5625			5.770				1.14	
-92	5.7500			6.8125			6.033				1.18	
-96	6.0000			7.0625			6.303				1.23	

1/ Torque values for these size bearings shall be specified in purchase orders.
2/ All dimensions to be after plating.
3/ Out-of- round tolerances: Bore: - 16 thru –49 +.0003, -52 thru –96 +.0005, -.0015.
Outer diameter: +.0010, -.0021.
4/ A radius giving approximately the same grip for staking the bearing in the housing will be acceptable.
5/ A radius giving approximately thru same fillet clearance will be acceptable.

TABLE II. BEARING ECCENTICITIES AND LOADS

MS DASH NO.	ECCENTRICITIES RADIAL INNER	ECCENTRICITIES RADIAL OUTER	ECCENTRICITIES LATERAL INNER	ECCENTRICITIES LATERAL OUTER	RADIAL LIMIT (LBS)	THRUST LIMIT RATING (LBS)	RADIAL LOAD RATING (LBS) FOR AVERAGE LIFE OF 10,000 COMPLETE 90≡ CYCLES 1/ 2/ CASE 1	RADIAL LOAD RATING (LBS) FOR AVERAGE LIFE OF 10,000 COMPLETE 90≡ CYCLES 1/ 2/ CASE II
-16					8085	3600	4260	3960
-21					9840	4400	4590	4290
-23					10500	4700	4650	4360
-25					11300	5000	4680	4420
-29	.0010	.0016	.0010	.0016	12700	5600	4760	4530
-33					14400	6400	4820	4630
-37					15800	7000	4880	4690
-47					24700	10900	6600	6390
-49					27500	12100	8150	7840
-52					28700	12600	8210	7880
-56					31200	13700	8240	7970
-60	.0012	.0016	.0010	.0016	33100	14600	8290	8010
-64					35000	15400	8350	8100
-68					41900	18400	10280	9900
-72					44300	19500	10330	10000
-76					46700	20500	10350	10050
-80					48200	21200	10360	10090
-84	.0014	.0018	.0012	.0018	50600	22300	10420	10120
-88					53000	23300	10480	10200
-92					55400	24400	10550	10240
-96	.0014	.0020	.0016	.0020	57700	25400	10520	10210

1/ Case 1 = load fixed with respect to outer ring.
Case 2 = load fixed with respect to inner ring.
2/ These ratings are for operations up to 250≡F. For operations up to 350≡F. The ratings shall be reduced by 20%.

REQUIREMENTS:

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1. MATERIAL: Rings: 52100 steel per AMS6440, or E52100 steel per FED-STD-66.
   Balls: 52100 steel per AMS6440 or 51100 steel per AMS6449, or E51100 or E52100 steel per FED-STD-66.
   Seals: Polytetrafluoroethylene per AMS3652 or Polytetrafluoroethylene sheet, glass fabric reinforced per AMS3666.
   Seal retainers: Any corrosion resistant steel.

2. LUBRICANT: MIL-G-81322 or MIL-G-23827, all bearings shall be packed with grease conforming to MIL-G-81322 unless otherwise specified. If MIL-G-23827 is required, add the letter "G" after the MS27642 dash number. MIL-G-23827 shall not be used for operation where temperatures exceed 250≡F.

3. HARDNESS: Heat treat rings and balls to rockwell "C" 60 to 66 and stabilize for operation at 250≡F.

4. SURFACE ROUGHNESS: Raceways and balls shall be 8 microinches Ra per ANSI/ASME B46.1.

5. PLATING: All external surfaces except bore, seals and seal retainers shall be zinc nickel in accordance with AMS 2417, Type 2, or cadmium plated in accordance with QQ-P-416, Type I, Class 2, with a thickness of .0003 to .0006 inches.

6. PART NUMBER: The part number shall be sequenced left to right with designations in the following sequence:

   EXAMPLE OF PART NUMBER:

   $^{+.0010}_{-.0005}$

   MS27642 – 16 S G = Bearing ball airframe, extra light duty, 1.000 inch bore -.0005 inch bore tolerance, .0001 to .0005 inch radial internal clearance, filled 80% minimum with MIL-G-23827 lubricant, 250≡F max.

   - G — Lubricant designator (see lubricant requirements)
   - S — Close bore tolerance designator (see Table I)
   - 16 — Dash number (see Table I)
   - MS27642 — Military Specification Sheet Number

7. Removable seals are required.

NOTES:

1. This specification takes precedence over any other documents referenced herein.

2. Referenced documents are the issues in effect at the date of invitation for bid.

3. The parts covered by MS20202 are cancelled. Use MS27642 for new design and replacement.

Custodians:
Army - AV
Air Force - 99
Navy - AS

Preparing activity:
Navy - AS

(Project 3110-1064)

Review activities:
Army – AR, AT, CR4, MI,
Air Force – 84
DLA-IS

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